



BIO 005 · HUMAN PHYSIOLOGY · YUBA COLLEGE

Week 1 Packet.

Foundations · Control · The Tools

1 What Physiology Is, and the Internal Environment
FOUNDATIONS

2 Control Systems and Feedback
HOMEOSTATIC CONTROL

3 The Quantitative Tools, and How We Know
METHOD AND MEASUREMENT

HOW TO USE THIS

Read it once, then close it.

Three parts, in the order the week teaches them. Part 1 is what the subject is and what it defends. Part 2 is the machinery that does the defending. Part 3 is the arithmetic and the evidence underneath both. Read a part, then close this and put what you can into the boxes on your note sheet. What you cannot recall is the gap, and the gap is what the videos are for.

Part 1, 2 and 3 are this packet's own parts, not Silverthorn's chapters. In the book, Week 1 is Chapter 1 plus the homeostasis and control pathway sections of Chapter 6. Where ATP comes up here it is being spent, not taught; the energy chemistry is Chapter 4, in the optional review week.

WHAT IS IN EACH PART, AND WHAT IT IS FOR

Element	What it is	How to use it
By the end	The competencies that part covers	Read them first. They are what the exam asks
Tables	Comparisons, side by side	Cover a column and recite it from memory
Sequences	A numbered process, step by step	Redraw the chain on blank paper without looking
Concept maps	A question with its branching answers	The question at the top is the one to ask yourself
Hold onto this	The idea the section turns on	If you remember one thing per section, this is it
Worked problems	A question, then the work in steps	Try it before you read the work. Getting it wrong first is the point

TWO COLORS

Work through the first pass in one color, before the videos. Then switch colors and add what the videos gave you that the reading did not. The gap between the two colors is the most useful thing on the page, so do not erase the first pass.

Dr. Sharilyn Rennie

PART 1

What Physiology Is, and the Internal Environment

FOUNDATIONS

What the subject is asking, the six levels it asks at, the fluid homeostasis is defending, why a steady state is not an equilibrium, and the six themes every mechanism this term sits on.

- BY THE END
1. Define physiology and place a given process at the correct level of organization from molecule to organism, then state the level at which that process is best explained.
 2. Predict how a change in the structure of a molecule, cell, tissue, or organ alters its function, using examples from two different organ systems.
 3. Name the body fluid compartments, give the fraction of body water in each, and say which one homeostasis defends.
 4. Define homeostasis, regulated variable and set point, and distinguish homeostasis from chemical equilibrium and from steady state.
 5. Name the six themes and say which two a given mechanism sits on.

What physiology is

THE WORDS THIS COURSE USES PRECISELY

Term	What it means
Physiology	The study of the normal functioning of a living organism and its parts, including the physical and chemical processes involved
Anatomy	The study of structure. Physiology asks what that structure does, and by what mechanism
Pathophysiology	What function looks like when a normal mechanism fails. Not a separate subject, the same mechanisms running wrong
Emergent property	A property of the whole system that none of its individual parts possesses. Blood pressure, consciousness, urine concentrating ability
Teleological answer	An answer that says what something is for . Useful shorthand, not an explanation
Mechanistic answer	An answer that says how something happens. This is what physiology means by an explanation, and what earns credit

THE ANSWER THAT EARNS THE MARK

A teleological answer says what something is for. A mechanistic answer says how it happens. Both can be true at once, and only one of them is physiology.

The six levels of organization

MOLECULES AT THE BOTTOM, THE ORGANISM AT THE TOP

- 1 Molecules**
Ions, proteins, lipids, nucleotides.
- 2 Cells**
The smallest unit that is alive.
- 3 Tissues**
Cells of one type working together.
- 4 Organs**
Several tissues doing one job.
- 5 Organ systems**
Organs sharing a function.
- 6 The organism**
Emergent properties appear only here.

WHY THE LEVEL YOU ANSWER AT MATTERS

Function at one level cannot be fully predicted from the level below it, which is why molecules alone do not explain a heartbeat. When a question asks you to place a process, place it where you would see it happening, then say which level actually explains it. Those are often not the same level.

Structure and function

THREE QUESTIONS THAT TURN A STRUCTURE INTO A PREDICTION

Ask this	What it gets you
What does this form allow?	Form sets what it can do
What is this wall separating?	Walls decide what mixes
How much surface is available?	Surface area sets exchange

1. Draw the structure as it normally is.
2. Draw it again with **one** feature changed.
3. Beside the second drawing, write what the function can no longer do.
4. Write why that loss **follows from** the change, rather than being a separate problem alongside it.
5. Do it a second time in a different organ system. Two systems, one shared reason, is what the competency asks for.

THE SENTENCE THAT GETS MARKED

The lost function has to follow from the structural change. If your answer would still make sense with the structure left intact, you have written two facts side by side instead of one explanation.

The internal environment

Homeostasis has to be defending something specific. This is what it is defending. Cells never touch the outside world, only the fluid immediately around them, and Claude Bernard called that fluid the milieu interieur.

READ IT FROM THE OUTSIDE IN

Compartment	What it is	How much
External environment	Everything the body is not. Air, water, food, temperature, whatever you are standing in	Outside
Gut lumen and airways	Open to the outside at both ends, so still external. A substance is only inside you once it has crossed an epithelium	Outside
Internal environment, the ECF	All fluid outside cells. This is what homeostasis defends, and this is the fluid a lab value is out of range in	About one third of total body water
Interstitial fluid	ECF sitting between cells. The fluid that actually bathes most tissue	About three quarters of the ECF
Plasma	The liquid part of blood. The only ECF compartment that circulates quickly, which is why it is the one you sample	About one quarter of the ECF
Intracellular fluid, ICF	Fluid inside cells, separated from the internal environment by one cell membrane and held at a completely different composition	About two thirds of total body water

Numbers worth carrying

- Total body water is about **60 percent** of body mass in an average adult male, closer to **50 percent** in an average adult female.
- Of that, roughly **two thirds is ICF** and **one third is ECF**.
- Of the ECF, roughly **three quarters is interstitial fluid** and **one quarter is plasma**.

FOOD SITTING IN YOUR STOMACH HAS NOT ENTERED YOU YET

The gut and the airways are drawn crossing the body wall on purpose. They sit inside the body outline and open to the outside at both ends, so they belong to the external environment. Mucus in the airway counts as outside the body for the same reason.

WHY THE SPLIT BETWEEN INSIDE AND OUTSIDE CELLS MATTERS

Sodium is high outside the cell and low inside. Potassium is the reverse. Those two gradients power most of what you study this term.

WHAT THIS MEANS IN THE CLINIC

Field	What it means	Example
Nursing	Where the extra fluid is sitting is what changes the things you see when you assess the patient	A patient can be visibly swollen, with fluid sitting in the interstitial space, and still be low on volume inside the vessels
Radiologic technology	Contrast goes into the blood first and then leaks out into the tissue, so when you scan decides which of those two you are looking at	An arterial phase scan catches contrast still inside the vessels. A delayed scan catches it after it has moved out into the interstitial space. Same injection, two different images, because the contrast moved compartments
Medicine	Before you pick an IV fluid you have to decide which fluid space you are trying to fill	A liter of isotonic saline stays largely in the extracellular fluid. A liter of 5 percent dextrose in water spreads across total body water, so much less of it stays in the vessels
Respiratory therapy	Mucus in the airway counts as outside the body, in the same way that food sitting in your gut is still outside you	A dry airway is a local humidity problem. It is not the same thing as whole body dehydration, and it is not fixed the same way

WORKED PROBLEM 1

An average adult male has a body mass of 70 kg. Work out his total body water, then the volume of each compartment, and say which one a blood sample reports on.

Total body water is about 60 percent of body mass in an average adult male. Two thirds of it is ICF, one third ECF. Three quarters of the ECF is interstitial fluid, one quarter is plasma. One kilogram of water is one liter.

THE WORK

1. Total body water: $70 \times 0.60 = 42 \text{ L}$.
2. Intracellular fluid: $42 \times 2/3 = 28 \text{ L}$. Most of the body's water is inside cells.
3. Extracellular fluid: $42 - 28 = 14 \text{ L}$. This is the internal environment.
4. Interstitial fluid: $14 \times 3/4 = 10.5 \text{ L}$. This is the fluid that bathes the cells.
5. Plasma: $14 \times 1/4 = 3.5 \text{ L}$. This is the only one that circulates quickly, so it is the one you sample.
6. If the same person were an average adult female, start from about 50 percent instead of 60 and every number below it changes with it.
7. **Draw:** a dashed box for the outside world, the body inside it, an ECF box inside that split into plasma and interstitial fluid, and the ICF below a line marked cell membrane.
8. **Carry:** you sample the smallest compartment and report on the one homeostasis is defending.

Homeostasis is not equilibrium

SIX DEFINITIONS THAT HAVE TO BE EXACT

Term	What it means
Homeostasis	The maintenance of a relatively stable internal environment. The word relatively is load bearing: the variable oscillates, it does not sit still
Dynamic steady state	Material and energy move continuously, but the amount in each compartment stays roughly constant. This is what homeostasis actually is
Equilibrium	A state in which no further net change occurs and no energy is required to maintain it. A cell at equilibrium with its surroundings is dead
Regulated variable	The quantity the body is actually holding in range, such as core temperature, plasma pH, or blood glucose
Set point	The target value the integrating center compares against. Set points move: they reset in fever, drift on a circadian cycle, and shift with acclimatization
Normal range	The band of values a variable is allowed to occupy. Usually defined to cover the middle 95 percent of a healthy population, so one healthy person in twenty falls outside it

EQUILIBRIUM AGAINST STEADY STATE

	Equilibrium	Steady state
Energy cost	Costs no energy to hold	Costs ATP every second
Concentrations	End up equal	Gradients are kept unequal
Is anything defended?	Nothing is being defended	Actively defended against change
Who reaches it	A dead cell, within hours	Only a living cell can hold it

THE STANDARD EXAMPLE

Extracellular sodium near 145 mM, intracellular sodium near 12 mM, held apart by a pump spending ATP every second of your life. Stop paying and the two numbers meet, which is equilibrium, which is death.

Why the line on the graph is wavy

- A regulated variable moves up and down inside its normal range, crossing the set point line rather than sitting on it.
- The variable is essentially **never** exactly at the set point.
- Negative feedback cannot prevent a change, only correct it, so oscillation is the normal result and not a sign of failure.
- A flat reading is worth a second look. Steady is not the same as still.

WHAT THIS MEANS IN THE CLINIC

Field	What it means	Example
Nursing	When blood flow drops, cells can no longer hold their gradients, and you see that in the patient before you see it in the labs	Cool mottled skin and a rising lactate are gradient failure becoming visible at the bedside
Radiologic technology	A nuclear medicine tracer only builds up where cells are still alive and still spending energy, so the picture is a map of which tissue is still working	Technetium sestamibi concentrates in heart muscle because living mitochondria maintain their charge gradient. Dead tissue cannot, so it takes up nothing, and that is what makes the cold spot on the scan
Medicine	Every gradient in the body costs energy to hold, and that bill never stops while the patient is alive	When perfusion stops, the sodium potassium pump stops with it, potassium leaks out of cells, and measured plasma potassium rises
Respiratory therapy	Breathing is muscle work, and that work is the price of keeping carbon dioxide moving out	A patient who is tiring can no longer afford that gradient, which is why carbon dioxide starts to climb as they wear out

WORKED PROBLEM 2

A cell is left in a solution until nothing about it changes any more and no energy is being spent on it. Name the state it has reached, say whether it is alive, and contrast it with what a living cell is doing with its sodium.

Equilibrium is a state in which no further net change occurs and no energy is required to maintain it. Extracellular sodium sits near 145 mM, intracellular sodium near 12 mM.

THE WORK

1. Nothing is changing and nothing is being spent. By definition that is **equilibrium**.
2. Homeostasis is not equilibrium. It is a **dynamic steady state**: material and energy move continuously, but the amount in each compartment stays roughly constant.
3. Equilibrium ends with concentrations equal and nothing being defended. Steady state keeps gradients unequal and defends them.
4. Equilibrium costs nothing to hold. Steady state costs ATP every second.
5. So the cell in the question is **dead**. Equilibrium is what a dead cell reaches within hours.
6. The living cell is holding extracellular sodium near 145 mM against intracellular sodium near 12 mM, and paying for that difference continuously.
7. **Draw**: two cells side by side, one with the sodium numbers apart and a pump drawn on it, one with the numbers equal and no pump.
8. **Carry**: steady is not still, and it is not free.

The six themes

USE THIS AS THE SPINE OF YOUR WEEK 1 NOTE SHEET, ONE BOX PER BRANCH

Physiology, six themes

Structure and function

- Form sets what it can do
- Walls decide what mixes
- Surface area sets exchange

Energy

- Gradients cost ATP
- Steady state is not free
- Metabolism pays for it

Information flow

- Chemical signals
- Electrical signals
- Signal, receptor, response

Homeostasis

- A variable held in range
- Sensor, integrator, effector
- Failure looks like disease

Compartmentation

- Inside against outside cells
- Plasma against interstitial
- Organelles inside the cell

Mass balance

- What comes in must leave
- Intake plus production
- Clearance removes it

TWO THEMES, EVERY TIME

Every mechanism this term sits on at least two of these. Take any mechanism from any week and name the two. If you cannot name two, you have memorized the mechanism without understanding where it came from, and naming them is the fastest check there is.

PART 2

Control Systems and Feedback

HOMEOSTATIC CONTROL

The four kinds of control, the seven boxes every reflex is made of, what a person looks like when each box fails, the three feedback patterns, and the three things that move a set point.

- BY THE END**
1. Distinguish a local control pathway from a long distance reflex pathway and classify a given response as one or the other.
 2. Diagram a negative feedback loop labeling stimulus, sensor, afferent path, integrating center, efferent path, effector and response, and trace body temperature or blood glucose through every step.
 3. Use the seven boxes as a differential and say what you would see in a person when each one fails.
 4. Distinguish negative from positive feedback by the direction of the response, give a physiological example of each, and explain why a positive feedback loop needs an outside event to end it.
 5. Explain anticipatory feedforward control and acclimatization, and identify which one is operating in a given scenario.

Four kinds of control

ASK WHAT IS DOING THE SENSING, AND HOW FAR THE SIGNAL HAS TO TRAVEL

How the body controls a variable

Local control

- One tissue only
- The cell that senses responds
- Low oxygen widens a vessel

Long distance control

- A reflex, whole body
- Nerves, hormones, or both
- Needs a path to travel

Tonic control

- Always switched on
- Turn it up or turn it down
- Vessel tone works this way

Antagonistic control

- Two systems, opposite jobs
- Faster and finer than one
- Insulin and glucagon

THE FOUR, DEFINED

Type	What it means
Local control	A response restricted to the tissue where the change was detected. The cell that senses the change also produces the response. Falling tissue oxygen dilates nearby arterioles
Long distance control	A reflex involving the nervous system, the endocrine system, or both, so a change in one place produces a response somewhere else
Tonic control	The signal is continuously on and the response is graded by changing its rate. Sympathetic tone in blood vessels works this way
Antagonistic control	Two systems push the variable in opposite directions. Insulin and glucagon on blood glucose. Faster and finer than a single system could be

WHAT DECIDES WHICH ONE YOU ARE LOOKING AT

Local control needs no signaling system at all, which makes it fast and limited. Reflex control needs a path, which makes it slower and general. Ask where the sensing happened and where the response appeared. If they are the same place, it is local.

WHAT THIS MEANS IN THE CLINIC

Field	What it means	Example
Nursing	Knowing whether the response is happening in one spot or across the whole body tells you what to expect when you do something about it	Warming one limb changes blood flow in that limb without changing the patient's core temperature
Radiologic technology	Contrast only lights up tissue that is actually getting blood, so the bright areas are a picture of which local vessels are open	A tumor lights up because its own vessels are dilated and leaky, a purely local change. Nothing whole body happened. The bright spot is local blood flow made visible
Medicine	Local control does not need nerves or hormones, so it keeps working even when the long distance systems have failed	In a transplanted heart the nerves are cut, and local metabolic control of the coronary vessels still works normally
Respiratory therapy	The lung sends blood to the parts that are getting air, and it does that locally, with no instruction from the brain	Low oxygen in one region narrows the vessels there, pushing blood toward regions that are actually being ventilated

The reflex loop

THE SEVEN STEPS OF A REFLEX CONTROL PATHWAY

- 1 Stimulus**
The change in the regulated variable that starts the loop.
- 2 Sensor, or receptor**
The structure that detects the change.
- 3 Afferent pathway**
Carries the signal toward the integrating center. Afferent means arriving.
- 4 Integrating center**
Compares the incoming signal with the set point and decides on a response.
- 5 Efferent pathway**
Carries the command outward. Efferent means exiting.
- 6 Effector**
The muscle or gland that does the physical work of the response.
- 7 Response**
The change produced, which feeds back on the sensor and closes the loop.

USE THE SEVEN BOXES AS A DIFFERENTIAL

Box	If this one breaks	What that looks like
Sensor	Nothing is detected	The body never learns that anything happened
Afferent pathway	Nothing arrives	The change was detected and the news never reached the integrating center
Integrating center	The comparison itself is wrong	Fever is a fault here, not in the sensors
Efferent pathway	The command is issued and never delivered	The response you are waiting for does not appear
Effector	The command arrives and nothing happens	Same picture as a broken efferent path, different address
Response	The loop never closes	The sensor keeps reporting the original stimulus

THE WHOLE SEVEN STEP SEQUENCE IS CALLED THE RESPONSE LOOP

Every reflex in this course is these seven boxes with different contents, so learn the shape once and then fill it in. When a regulated variable is out of range, the fault is in one of the boxes, and this list is the first thing to write when your patient file goes wrong.

WHAT THIS MEANS IN THE CLINIC

Field	What it means	Example
Nursing	Name which part of the loop is broken and you know which sign to watch for, and which sign is never going to show up	In a patient on a beta blocker, do not wait for a rising heart rate to tell you they are bleeding. Watch the other signs
Radiologic technology	Reactions on the table fall into two groups depending on which pathway is firing, and the two look nothing alike	A patient who goes pale, sweaty and slow pulsed after an injection is a vagal reaction: the efferent nerve output dropped the heart rate. Anaphylaxis drives the heart rate up instead. Same table, opposite pathways
Medicine	The seven boxes give you a list of suspects, and each broken box produces its own picture	A patient who cannot raise their heart rate while bleeding may have an intact sensor and a blocked efferent path, which is exactly what a beta blocker does
Respiratory therapy	Nearly everything you do starts with a chemoreceptor noticing something, and that is the sensor box of the loop	If the sensor is blunted, the drive to breathe falls even though the stimulus telling it to rise is still there

WORKED PROBLEM 1

A patient on a beta blocker is bleeding. Their heart rate does not rise. Name the box that is broken, say which boxes are intact, and say which sign you should therefore not be waiting for.

Beta blockers act on the efferent side of the cardiovascular reflex, not on the sensors.

THE WORK

1. Write the seven boxes out before you decide anything: stimulus, sensor, afferent path, integrating center, efferent path, effector, response.
2. **Stimulus.** Blood volume is falling. Real, and present.
3. **Sensor and afferent path.** Intact. The change is detected and the signal arrives.
4. **Integrating center.** Intact. It compares and issues a command to raise heart rate.
5. **Efferent path. Blocked.** The command is issued and the heart never receives it.
6. So the response never appears, and heart rate stays flat while the patient goes on bleeding.
7. The flat heart rate is not evidence that nothing is wrong. It is evidence that one box is out of service.
8. **Draw:** the seven boxes in a row, with a cross through the efferent arrow and a tick over the first four.
9. **Carry:** naming the broken box tells you which sign to watch and, just as usefully, which sign is never going to show up.

Feedback types

THE THREE PATTERNS, COMPARED ON THE ONLY THREE THINGS THAT SEPARATE THEM

Pattern	What the response does	What it is for	How it stops
Negative feedback	Opposes the stimulus	Holding a variable near a set point	Shuts itself off as the variable comes back
Positive feedback	Reinforces the stimulus	Driving a process to completion	Needs an outside event to end it
Feedforward	Acts before the variable has changed at all	Blunting a change you can see coming	Ends when the anticipated load arrives, or does not

Three patterns cover almost every loop in this course. They differ in one thing only: what the response does to the stimulus that triggered it.

NEGATIVE AND POSITIVE ARE DIRECTIONS, NOT VERDICTS

Positive feedback is not the body malfunctioning and negative feedback is not the body being cautious. Negative means the response subtracts from the stimulus and positive means it adds to it. That is the whole distinction, and it says nothing about whether the outcome is good for the patient.

Negative feedback

- The response **opposes** the stimulus that produced it.
- Stabilizing. It shuts itself off as the variable returns toward the set point.
- It cannot prevent the initial change, only correct it, which is why regulated variables oscillate.

THERMOREGULATION, WORKED ALL THE WAY THROUGH

- 1 Core temperature falls**
A cold room. This is the stimulus.
- 2 Thermoreceptors detect it**
In the skin and in the hypothalamus. These are the sensors.
- 3 The hypothalamus compares it with the set point**
Near 37 C (98.6 F). This is the integrating center.
- 4 Sympathetic and motor output leave**
These are the efferent signals.
- 5 Vessels constrict and muscle shivers**
These are the effectors.
- 6 Heat production rises and heat loss is cut**
Core temperature returns toward the set point, and the sensors stop reporting cold.

REPRODUCE THIS ONE FROM MEMORY

Draw the seven boxes, then run body temperature through them with the real anatomy written in each box. Then do it again with blood glucose. Those two are what the competency names.

WHAT THIS MEANS IN THE CLINIC

Field	What it means	Example
Nursing	A body that is still fixing the problem looks different from one that has run out of room, and telling those apart is most of assessment	A brief drop on standing that recovers is the loop working. A drop that stays down is the loop failing
Radiologic technology	Every time you sit a patient up or lay them flat you are running this loop live and watching what it does	Sitting a patient up from a supine scan drops their pressure for a few seconds before the reflex catches it. In an older or dehydrated patient the catch is slow, and that is the moment they go down. It is why you move them in stages and stay with them
Medicine	Negative feedback cannot stop a change from happening. It can only pull it back, so the value always moves first	Blood pressure dips on standing before the reflex catches it. The dip is normal. Failing to recover from it is not
Respiratory therapy	Breathing harder is the body's negative feedback answer to carbon dioxide going up	Rising carbon dioxide normally drives ventilation up. If the carbon dioxide is high and ventilation is not rising, the loop is broken somewhere

Positive feedback

- The response **reinforces** the stimulus, pushing the variable further from where it started.
- Destabilizing by design, so it needs an event **outside the loop** to stop it.
- Oxytocin in labor, stopped by delivery. The clotting cascade, stopped when the vessel seals. Sodium entry during an action potential, stopped by channel inactivation.
- Positive feedback with no off switch describes dying: uncontrolled hemorrhage and runaway hyperkalemia both work this way.

THE ENDING IS THE PART PEOPLE LEAVE OUT

For every positive feedback example you write down, write the outside event that shuts it off next to it. An answer that names the amplification and not the ending is half an answer.

Feedforward control

- The body responds to a signal that **predicts** a change, before the regulated variable has moved.
- Salivating and secreting gastric acid at the smell of food. Heart rate rising in the seconds before exercise begins.
- Feedforward buys time. It cannot correct an error, because at that moment there is no error.

THE DISTINCTION, ONE LINE EACH

Negative feedback reacts to what already happened. Feedforward reacts to what is about to happen.
Positive feedback commits to what is happening now.

Set points move**THREE THINGS THAT MOVE A SET POINT, ON THREE DIFFERENT TIMESCALES**

What moves it	Timescale	What you see
Fever	Hours	A deliberately raised set point, not a broken thermostat. You shiver and feel cold while your temperature is already climbing, because the body reads 37 C (98.6 F) as too cold against the new target
Circadian rhythm	A daily cycle	Core temperature, cortisol and blood pressure all have a normal range that depends on the hour
Acclimatization	Days to weeks	Hematocrit rises at altitude. Sweat sodium falls in a hot climate

FEEDFORWARD AGAINST ACCLIMATIZATION, THE QUESTION THAT SEPARATES THEM

Ask how long the change took. Feedforward happens in seconds, before the load arrives, and it ends when the load arrives or does not. Acclimatization takes days to weeks and moves the target itself.

WHAT THIS MEANS IN THE CLINIC

Field	What it means	Example
Nursing	Cool a patient who is shivering and you are fighting the thermostat instead of turning it down	Shivering generates heat, so external cooling during the rise can drive the temperature higher rather than lower
Radiologic technology	Scanner rooms are kept cold for the machine, and the patient's thermostat reacts to that whether you want it to or not	A cold, uncovered patient shivers, and shivering is movement, and movement is motion artifact. The warm blanket is an image quality tool, not just a comfort measure
Medicine	In a fever the thermostat is not broken. It has been turned up, and the body is working hard to reach the new number	The patient shivers while their temperature is already climbing, because the body reads its current temperature as too cold against the new target
Respiratory therapy	If carbon dioxide stays high for long enough, the sensor stops treating it as an emergency and settles on a new normal	In long standing carbon dioxide retention the body adapts to a higher normal, so the textbook number is no longer that patient's target

WORKED PROBLEM 2

A patient with a fever is shivering hard. Their temperature is already climbing. Explain why they feel cold, and say why external cooling at this moment can make the number go up rather than down.

Fever is a raised set point, not a broken thermostat. Shivering is an effector response and it generates heat.

THE WORK

1. The integrating center has been given a **new, higher set point**. Nothing is broken.
2. The sensors report the current core temperature honestly. The comparison is now made against the new target.
3. Against that new target, 37 C (98.6 F) reads as **too cold**, so the loop issues a warming command.
4. The effectors do what they are told: vessels constrict and muscle shivers. That is why the patient feels cold while the number rises.
5. Cool them externally now and you have not moved the set point. You have widened the gap the loop is trying to close.
6. The loop answers by shivering harder, and shivering generates heat, so the temperature can rise rather than fall.
7. **Draw:** a set point line that steps upward, with the temperature line climbing to meet it, and mark the interval where the patient feels cold.
8. **Carry:** before you fight a response, check whether the set point moved. Cool a shivering patient and you are fighting the thermostat instead of turning it down.

WORKED PROBLEM 3

Two people are described. One is standing on a start line and their heart rate rises in the seconds before the race begins. The other has been at altitude for three weeks and their hematocrit has risen. Say which is feedforward and which is acclimatization, and give the one question that separates them.

Feedforward acts on a signal that predicts a change, before the regulated variable has moved. Acclimatization resets a set point over days to weeks.

THE WORK

1. **Ask how long the change took.** That is the question that separates them, and it settles both cases at once.
2. **The start line.** Heart rate rises in **seconds**, and it rises before the exercise has changed anything the body is regulating.
3. Nothing is out of range yet, so there is no error to correct. The body is acting on a prediction. That is **feedforward**.
4. It ends when the anticipated load arrives, or does not. Feedforward buys time, it cannot correct an error.
5. **Altitude.** Three weeks is not seconds. Nothing is being anticipated, and the change has become the new normal.
6. A set point has been reset over days to weeks. That is **acclimatization**. Sweat sodium falling in a hot climate is the same thing in a different system.
7. **Draw:** one timeline in seconds with the response drawn before the load, and one timeline in weeks with the target line itself moving.
8. **Carry:** feedforward acts before a change and leaves the set point alone. Acclimatization moves the set point and takes weeks to do it.

PART 3

The Quantitative Tools, and How We Know

METHOD AND MEASUREMENT

Three equations, the units they are written in, how to build and read a graph, the words an experiment is described in, and why one flagged value on a panel is not an answer.

- BY THE END
1. Apply the mass balance equation to a solute or to body water and calculate the intake, production and output combination that holds the amount in the body constant.
 2. Convert among the units used in physiology including molarity, osmolarity, milliequivalents, mmHg, liters per minute and percent solutions.
 3. Construct a labeled graph with the independent variable on the x axis, and read slope, direction and trend from a physiological data set.
 4. Identify the hypothesis, independent variable, dependent variable and control condition in a physiology experiment and state what the control rules out.
 5. Distinguish random from systematic error and explain why physiological measurements are repeated and averaged.

Three equations, and what each one is for

THE QUANTITATIVE TOOLS

Equation	Units	What it says
Total body load = intake + production – excretion – metabolism	An amount, not a concentration	A concentration can change for two entirely different reasons. Either the amount of substance changed, or the volume it is dissolved in changed
Mass flow = concentration × volume flow	mg/min = mg/mL × mL/min	Delivery is not the same as concentration. A normal oxygen concentration with a collapsed cardiac output still starves the tissue
Clearance = rate of removal ÷ plasma concentration	mL/min, a volume per unit time, not an amount per unit time	The volume of plasma completely cleared of a substance per minute. No real volume is ever fully cleared

LOAD IS NOT A CONCENTRATION

Load is the total amount of a substance in the body. Never conclude that the amount changed without first ruling out that the volume changed. A rising plasma sodium can mean sodium was gained or water was lost, and the treatments are opposite.

- Mass flow is why shock kills people whose blood looks normal. The concentration is fine and the delivery is not.
- Expressing removal as a clearance makes it independent of how much of the substance happens to be present, which is why it is the number used to set drug doses.

WHAT THIS MEANS IN THE CLINIC

Field	What it means	Example
Nursing	Intake and output charting is just what went in minus what came out. Record it loosely and the problem hides in the gap	A patient in positive balance across three shifts has gained fluid, and the daily weight usually shows it before anything else does
Radiologic technology	Contrast follows the same rule as everything else. What you put in stays in the patient until the kidney clears it out	A patient who had contrast yesterday has not finished clearing it today. A repeat study too soon adds to a load that is still there, which is why prior studies get checked before the next injection
Medicine	A number can rise because there is more of the substance, or because there is less water, and those two need opposite treatments	A rising sodium can mean sodium was gained or water was lost. Treating the wrong one moves the patient in the wrong direction
Respiratory therapy	Extra fluid ends up in the lung, and water in the lung gets in the way of gas crossing over	A patient several liters positive becomes harder to oxygenate, and the chest film tends to follow the balance sheet

Mass balance

- Mass balance is **in minus out**, and nothing more.
- What comes in must leave. Intake plus production on one side, excretion plus metabolism on the other.
- If the two sides are equal, the total body load is constant. That is what steady state means for a substance.
- If one arrow gets bigger and nothing else changes, the load moves. That is the whole of the prediction.

WORKED PROBLEM 1

A solute is at steady state in the body. Intake is 150 mmol per day. The body produces 30 mmol per day of it and metabolism destroys 20 mmol per day. How much must be excreted per day to hold the total body load constant? Then say what happens if excretion falls to 120 mmol per day and nothing else changes.

Total body load = intake + production – excretion – metabolism. Numbers chosen to practice the equation, not laboratory values.

THE WORK

1. For the load to be constant, the change in load per day must be zero.
2. $0 = \text{intake} + \text{production} - \text{excretion} - \text{metabolism}$.
3. $0 = 150 + 30 - \text{excretion} - 20$.
4. Everything in must be matched by everything out: $150 + 30 = 180$ in, and 20 of that is destroyed by metabolism.
5. $\text{Excretion} = 180 - 20 = \mathbf{160 \text{ mmol per day}}$.
6. Now drop excretion to 120 and change nothing else: $150 + 30 - 120 - 20 = \mathbf{+40 \text{ mmol per day}}$.
7. The body gains 40 mmol of this solute every day, and it keeps gaining until one of the other three arrows moves.
8. **Draw:** a person with two arrows in and two arrows out, numbers on each, and the balance written underneath.
9. **Carry:** a load that is rising is not a mystery. One of four arrows moved, and the equation tells you which ones to check.

WORKED PROBLEM 2

A patient's plasma sodium has risen since yesterday. Using total body load, list the two ways that number could have risen and say what you have to rule out before you treat it.

A concentration is an amount divided by a volume. Either term can change.

THE WORK

1. Plasma sodium is a **concentration**: sodium amount divided by the volume it is dissolved in.
2. A concentration rises if the numerator rises. Sodium was gained, so intake exceeded excretion.
3. A concentration also rises if the denominator falls. Water was lost, and the sodium that was already there is now in less water.
4. Nothing about the number itself tells you which happened. Both produce the same reading.
5. The two situations need **opposite** treatments, so treating the wrong one moves the patient in the wrong direction.
6. What settles it is the balance sheet: what went in, what came out, and what the daily weight did.
7. **Draw:** the same number written as a fraction twice, once with the top raised, once with the bottom lowered.
8. **Carry:** never conclude the amount changed without ruling out that the volume changed.

Units and unit conversion

THE MATH UNIT 1 ASSUMES YOU ALREADY HAVE

Skill	Where it turns up in this week
Metric prefixes and unit conversion	90 mg/dL and 0.9 mg/mL are the same quantity written two ways. Convert first, then multiply
Counting particles, not molecules	Molarity counts molecules. Osmolarity counts particles. They are not the same number
Rates, and what per means	Mass flow is mg per minute. Clearance is mL per minute. The word per is doing the work
Reading a graph	Every data set in the lab arrives as a graph before it arrives as a conclusion
What a log scale does	pH is on the list of things this week assumes you already have, and pH is a log scale
In minus out	The mass balance equation is this and nothing more
Charge, and why mEq is not mM	Charge is what a milliequivalent counts, so the two numbers are not interchangeable
Sanity checking an answer	Carry the units through the arithmetic. A wrong answer usually announces itself in the units before you spot it in the number

- The units this course asks you to convert among are **molarity, osmolarity, milliequivalents, mmHg, liters per minute and percent solutions**.
- Build yourself a conversion map: one box per unit, arrows between the ones you can convert between, the conversion written along each arrow.
- Mark the two you keep getting wrong. Those are the two to practice, and nobody else can tell you which two they are.

WORKED PROBLEM 3

Plasma glucose is 90 mg/dL and renal plasma flow is 600 mL/min. How much glucose is delivered to the kidney per minute?

Mass flow = concentration $\times$ volume flow. The units have to match before you multiply.

THE WORK

1. The concentration is per deciliter. The flow is per minute in milliliters. They do not match, so convert first.
2. There are 100 mL in a dL, so $90 \text{ mg/dL} = 90 \div 100 = \mathbf{0.9 \text{ mg/mL}}$.
3. Now the units line up: $\text{mg/mL} \times \text{mL/min}$ leaves mg/min .
4. Mass flow = $0.9 \text{ mg/mL} \times 600 \text{ mL/min} = \mathbf{540 \text{ mg/min}}$.
5. Essentially none of it appears in the urine.
6. So something is moving 540 mg of glucose per minute back across an epithelium, against its gradient. That transporter is the subject of Week 14.
7. **Draw:** the unit cancellation written out, mL over mL crossed through, so you can see why the answer has to be mg/min.
8. **Carry:** convert, then multiply, then check that the units of your answer are the units the question asked for.

Graphing and data interpretation

1. Put the **independent variable on the x axis**. That is what the experimenter deliberately changed.
2. Put the **dependent variable on the y axis**. That is what was measured in response.
3. Label both axes with the quantity and its unit. An unlabeled axis is not a graph.
4. Read direction first: is the line going up or down as x increases?
5. Then read slope: where is it steep, where is it flat?
6. Then read trend: what is the shape doing overall, across the whole range?

THREE SHAPES WORTH BEING ABLE TO SKETCH

Shape	How to read it
A straight line	The dependent variable changes by the same amount for every step in the independent variable, right across the range
A curve that rises then flattens	The response climbs, then stops climbing. Something is running out, or a ceiling has been reached
A curve that rises then falls	The relationship reverses past a peak. More of the independent variable stops helping and starts hurting

WHAT A STEEP SLOPE AND A FLAT SLOPE ARE TELLING YOU

A steep slope means a small change in the independent variable produces a large change in the measured one, so the system is sensitive there. A flat slope means the measured variable barely moves, so the system is insensitive there, or already at its limit. Mark both on every graph you sketch.

- Figure 4 of the lecture notes, the regulated variable oscillating inside a shaded normal range around a dashed set point, is a physiological graph. Time is on the x axis because time is what is being allowed to run.
- A flat reading on that graph does not by itself tell you the system is idle. Write down what it does and does not tell you.

How we know any of this

THE WORDS AN EXPERIMENT IS DESCRIBED IN

Term	What it means
Independent variable	What the experimenter deliberately changes
Dependent variable	What is measured in response to that change
Control	An otherwise identical condition in which the independent variable is not changed
Placebo	An inactive treatment used to separate the effect of a substance from the effect of being treated
Blind study	The participant does not know which group they are in
Double blind study	Neither the participant nor the person measuring the outcome knows
Reference range	The band of values covering the middle 95 percent of a healthy reference population. Not a definition of health
Biological variability	Real differences between healthy people. Not measurement error, and not something to be averaged away

WHAT THE CONTROL RULES OUT

The control condition is identical in every respect except the one you changed. Everything the two conditions share is therefore ruled out as an explanation of the difference, which leaves the independent variable. Write that as a full sentence naming what your control rules out, because that is the part that gets marked.

Random error against systematic error

TWO KINDS OF ERROR, AND ONLY ONE OF THEM AVERAGES AWAY

	Random error	Systematic error
On a target	Shots scattered all around the center	Shots grouped tightly, but off center
In five repeats	The dots scatter around the right value	The dots sit together in the wrong place
Does repeating and averaging fix it?	Yes. The scatter shrinks as you average more measurements	No. Averaging a biased measurement gives you a precise wrong answer
How you detect it	Spread across repeats of the same measurement	Comparison against a known standard or a second method

- This is why physiological measurements are repeated and averaged: repetition is the defense against random error.
- It is no defense at all against systematic error, which is why the instrument has to be checked separately.
- **Biological variability is neither.** It is real difference between healthy people, and averaging it away throws information out.

WHAT THIS MEANS IN THE CLINIC

Field	What it means	Example
Nursing	One number on its own tells you very little. What that number did since last time tells you a lot	A creatinine at the top of normal that doubled since yesterday is a bigger problem than one that is mildly high and unchanged
Radiologic technology	You check lab numbers before giving contrast, and a number outside the normal range does not automatically mean stop	A creatinine sitting slightly high and unchanged for a year is a different situation from one that doubled this week. Both print with the same flag on the report. Only one of them changes what happens next
Medicine	Normal ranges are built to cover most healthy people, not every healthy person, so about one healthy person in twenty gets flagged on any given test	Run a twenty analyte panel on a completely healthy person and something comes back flagged more often than not
Respiratory therapy	A blood gas is one photo of something that is still moving	The same value on the way up and on the way down mean opposite things about where the patient is heading

WORKED PROBLEM 4

A completely healthy person has a twenty analyte panel run. One result comes back flagged. How surprised should you be, and what does the flagged value actually justify doing?

A reference range is defined to cover the middle 95 percent of a healthy reference population.

THE WORK

1. If the range covers the middle 95 percent of healthy people, then **5 percent** of healthy people fall outside it on any one test.
2. 5 percent is one in twenty. So each individual test flags **one healthy person in twenty**, by construction, with nothing wrong.
3. Run twenty independent tests on one healthy person and you have twenty chances for that to happen.
4. The odds of at least one flag are close to even. Something coming back flagged is the ordinary outcome, not the surprising one.
5. So a flagged value is a **reason to look further, not an answer**.
6. What to look at next is the trend. A creatinine at the top of normal that doubled since yesterday is a bigger problem than one that is mildly high and unchanged. Both print with the same flag.
7. **Draw:** a normal curve with the middle 95 percent shaded and the two tails left white, then twenty tick marks for twenty tests.
8. **Carry:** one number on its own tells you very little. What that number did since last time tells you a lot, and the story only shows up across the whole set.